

Question number		
6 (a)	The only correct answer is C <i>A is incorrect because enzymes are organic molecules that decrease activation energy</i> <i>B is incorrect because enzymes are organic molecules</i> <i>D is incorrect because enzymes increase activation energy</i>	(1)

Question number	Answer	Additional guidance	Mark
6(b)(i)	log / logarithmic / $\log_{10}$ / log to the base 10 (scale)	DO NOT ACCEPT $\ln$ / $\log_e$ / natural log	(1)

Question number	Answer	Mark
6(b)(ii)	The only correct answer is C <i>A is incorrect because both level of salt and wave action are abiotic</i> <i>B is incorrect because wave action is abiotic</i> <i>D is incorrect because level of salt is abiotic</i>	(1)

Question number	Answer	Additional guidance	Mark
6(b)(iii)	An explanation that includes three of the following points: - enzymes needed for the metabolism (of the fish) (1) - because enzyme activity is not sufficient at both higher and lower temperatures (than zone of tolerance) (1) - because at lower temperatures there will not be enough kinetic energy for {collisions / enzyme-substrate complexes to be formed} (1) - because at higher temperatures {the enzymes denature / active site changes shape} (1)	PIECE TOGETHER ACCEPT converse ACCEPT enzymes do not work fast enough / no enzyme activity / in zone of tolerance enzyme <u>activity</u> is at an optimum DO NOT ACCEPT start to denature	(3)

Question number	Answer	Additional guidance	Mark
6(c)(i)	- values for 20 °C and 30 °C read from the graph (1) - numbers substituted into the equation and evaluated (1) 1.5 / 1.6 / 1.56 / 1.62 with no units (1)	{2.6 / 2.7 / 2.8} and 4.2 $4.2 \div 2.6 = 1.615384615$ $4.2 \div 2.7 = 1.555555555556$ $4.2 \div 2.8 = 1.5$ answer to 1 or 2 decimal places Bald answer of 1.5 / 1.6 / 1.56 / 1.62 with no units = 3 marks Bald answer of 1.556 / 1.615 or more correctly rounded dps = 2 marks Bald answer of {2.6 / 2.7 / 2.8} and 4.2 = 1 mark	(3)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	An explanation that includes three of the following points: • as storage time increases the {activity / optimum temperature (of the enzyme)} decreases (1) • the active site changes shape (due to storage time) (1) • because the bonds break (due to more energy) (1) • as more energy causes more vibration between the R groups (1) • the longer the active site is distorted the less likely it is to return to normal shape (1)	ACCEPT tolerance to temperature decreases ACCEPT enzyme {changes shape / changes structure / is denatured} DO NOT ACCEPT any named bond not found in an enzyme	(3)